

Source-Cyclic Spectral Quotients

A Type-Correct Reduction of Matrix-Valued Left-Multiplication Orbits

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Abstract

RH-192 proves that the full Frobenius operator $\mathcal{L}_A : X \mapsto AX$ is m copies of A and therefore cannot have the rank-one or rank-four ambient Riesz shells sought by the temporal packet. This paper constructs the exact reduced state selected by one physical matrix source:

$$\mathcal{K}_S = \text{span}\{S, AS, A^2S, \dots\}.$$

The space is $\mathcal{L}_A$ -invariant, cyclic, and has dimension

$$\dim \mathcal{K}_S = \deg \mu_{A,S} \leq \deg \mu_A \leq n,$$

independently of the source width m . The restriction has minimal and characteristic polynomial $\mu_{A,S}$. Every right temporal packet built from the source orbit lies exactly in $\mathcal{K}_S$.

For an observation seed O^* of the same matrix type, orthogonal restriction to $\mathcal{K}_S$ preserves every moment $h_j = \langle O^*, A^j S \rangle_F$ and hence the complete scalar resolvent germ at infinity. A 140-case complex Arnoldi audit verifies invariance, intertwining, packet inclusion, and moment preservation with zero failures.

The result removes the m -fold multiplicity obstruction at the correct source-relative type. It does not assert that $\dim \mathcal{K}_S$ is uniformly small in the physical limit. The next task is to determine whether the local length-four temporal roots match actual source-observable spectral modes of the base operator.

1 From an ambient no-go to a cyclic state

Let $A \in \mathbb{C}^{n \times n}$, let $S \in \mathbb{C}^{n \times m}$ be nonzero, and equip $\mathcal{H}_F = \mathbb{C}^{n \times m}$ with the Frobenius inner product. Define $\mathcal{L}_A X = AX$. RH-192 identifies $\mathcal{L}_A$ with $I_m \otimes A$, so its full Riesz multiplicities are multiples of m [1].

The physical experiment, however, never initializes an arbitrary element of $\mathcal{H}_F$. It initializes the single vector S and observes its orbit. The smallest invariant state containing that experiment is therefore a cyclic subspace, not the complete direct sum.

2 The source annihilator

Define the ideal

$$\mathcal{I}_{A,S} = \{p \in \mathbb{C}[z] : p(A)S = 0\}. \quad (1)$$

It is nonzero because the characteristic polynomial of A belongs to it. Since $\mathbb{C}[z]$ is a principal ideal domain, there is a unique monic generator, denoted $\mu_{A,S}$.

Proposition 2.1 (Divisibility). *The source minimal polynomial $\mu_{A,S}$ divides the ordinary minimal polynomial μ_A and therefore*

$$\deg \mu_{A,S} \leq \deg \mu_A \leq n. \quad (2)$$

Proof. Since $\mu_A(A) = 0$, one has $\mu_A(A)S = 0$, so $\mu_A \in \mathcal{I}_{A,S} = (\mu_{A,S})$. $\square$

The polynomial depends on the complete matrix source. Equivalently it is the least common multiple of the vector minimal polynomials of the columns of S . The columns can activate more modes, but they do not create m independent copies when the entire matrix is treated as one source vector.

3 Exact cyclic restriction

Set

$$\mathcal{K}_S = \text{span}\{A^j S : j \geq 0\} \subset \mathcal{H}_F. \quad (3)$$

Theorem 3.1 (Source-cyclic realization). *The space $\mathcal{K}_S$ is invariant under $\mathcal{L}_A$ and*

$$\dim \mathcal{K}_S = \deg \mu_{A,S} \leq n. \quad (4)$$

The restricted operator $T_S = \mathcal{L}_A|_{\mathcal{K}_S}$ is cyclic with cyclic vector S . Its minimal and characteristic polynomials both equal $\mu_{A,S}$.

Proof. Invariance follows from

$$\mathcal{L}_A(A^j S) = A^{j+1} S.$$

If $d = \deg \mu_{A,S}$, then $S, AS, \dots, A^{d-1}S$ are linearly independent; otherwise a polynomial of degree below d would annihilate S . The relation $\mu_{A,S}(A)S = 0$ expresses every later power in their span, proving the dimension statement. The restriction is generated by S , so it is a cyclic d -dimensional operator. For a cyclic operator the minimal polynomial has degree d and equals the characteristic polynomial. $\square$

The theorem is exact and contains no numerical-rank threshold. In a floating implementation, numerical cyclic dimension can be delicate because long power sequences become ill-conditioned. This affects computation, not the algebraic state type.

4 Coordinate realization

Let

$$Q : \mathbb{C}^d \longrightarrow \mathcal{K}_S \quad (5)$$

be any Frobenius isometry onto the cyclic space. Define

$$T = Q^* \mathcal{L}_A Q, \quad b = Q^* S. \quad (6)$$

Proposition 4.1 (Intertwining). *One has*

$$\mathcal{L}_A Q = QT, \quad A^j S = QT^j b \quad (j \geq 0). \quad (7)$$

The spectrum and algebraic multiplicities of T are those of the source-relative cyclic restriction, with no forced factor m .

Proof. Because $\text{Ran } Q$ is invariant, $\mathcal{L}_A Q$ lies in $\text{Ran } Q$ and equals $QQ^* \mathcal{L}_A Q = QT$. Induction gives the orbit identity. $\square$

If the source is cyclic for A in the matrix sense, then $d = n$. This still reduces the ambient dimension from nm to n , which is decisive at the physical widths 64–256.

5 Exact inclusion of temporal packets

For indices $t_1, \dots, t_L$, let the unnormalized temporal synthesis be

$$J = [A^{t_1} S, \dots, A^{t_L} S]. \quad (8)$$

Column normalization, polar orthogonalization, and invertible right coordinate changes do not alter its range.

Corollary 5.1 (Packet inclusion). *Every right temporal subspace constructed from the source orbit is contained in $\mathcal{K}_S$. In reduced coordinates,*

$$J = Q[T^{t_1} b, \dots, T^{t_L} b]. \quad (9)$$

This includes the RH-182 orthogonal packet and the right half of the RH-185 balanced bi-Krylov packet. The left temporal vectors need not lie in the same cyclic space, but they restrict to bounded coordinate functionals $Q^* W$. If $W^* V = I$ and $V \subset \mathcal{K}_S$, then

$$(Q^* W)^* (Q^* V) = W^* Q Q^* V = W^* V = I. \quad (10)$$

Thus biorthogonality is preserved when the left frame is restricted to the source-cyclic state.

6 Observation moments

Let $O^* \in \mathcal{H}_F$ be an observation seed and define the scalar moments

$$h_j = \langle O^*, A^j S \rangle_F. \quad (11)$$

Write $c = Q^* O^*$.

Theorem 6.1 (Exact moment realization). *For every $j \geq 0$,*

$$h_j = c^* T^j b. \quad (12)$$

For $|z| > \rho(T)$,

$$\langle O^*, (zI - \mathcal{L}_A)^{-1} S \rangle_F = c^* (zI - T)^{-1} b = \sum_{j \geq 0} \frac{h_j}{z^{j+1}}. \quad (13)$$

Proof. Insert $A^j S = QT^j b$ and use that Q is an isometry. The resolvent formula follows from the convergent Neumann series. $\square$

The source-cyclic restriction therefore loses no scalar input-output data. Modes absent from T are exactly modes not excited by the matrix source. A further observability quotient can remove modes whose residues vanish.

7 Riesz projectors in the cyclic state

For a contour Γ in the resolvent set of T , let $P_\Gamma(T)$ be its Riesz projector. Intertwining and holomorphic functional calculus give

$$P_\Gamma(\mathcal{L}_A)Q = QP_\Gamma(T). \quad (14)$$

The rank on $\mathcal{K}_S$ is therefore $\text{rank } P_\Gamma(T)$, not $m \text{rank } P_\Gamma(A)$. For one simple source-excited base eigenvalue it can equal one.

There is no contradiction with RH-192. The full ambient eigenspace has m copies, while the cyclic orbit chooses at most one polynomially generated direction through that eigenspace. The remaining $m - 1$ directions are not reachable from the single matrix seed.

8 Finite Arnoldi audit

The implementation performs Frobenius Arnoldi on complex nonnormal matrices of dimensions 2 through 8 and source widths 1 through 4. Five independent trials per pair give 140 cases. Each matrix is similar to a diagonal matrix with distinct eigenvalues, and each random source activates all modes.

The audit checks:

1. Frobenius orthogonality of the computed basis;
2. closure and the intertwining defect $\mathcal{L}_A Q - QT$;

3. inclusion of the first temporal powers;
4. equality of $2n$ full and reduced moments for an independent observation seed.

All 140 cases pass the declared tolerances. This audit tests the implementation of exact identities; it is not evidence for a uniform physical numerical rank.

9 What is and is not gained

The gain is structural:

$$nm\text{-dimensional repeated spectrum} \longrightarrow d \leq n\text{-dimensional source-cyclic spectrum.}$$

The temporal packet is now in the correct invariant state, and its left observation functionals restrict without breaking biorthogonality.

What remains open is equally important. The physical value of d may grow with refinement. A four-vector temporal window does not imply $d = 4$. Moreover, the cyclic space is source-dependent. Gate A eventually needs a selection that is canonical enough to transport between levels and to feed an intrinsic cloud ledger.

The immediate falsifiable question is finite: do the four temporal roots in the surviving $\sigma = 0.01$ windows lie near actual eigenvalues activated by the source and seen by the observation? RH-194 performs that audit.

References

- [1] Liang Wang. The frobenius left-multiplication obstruction, 2026. Prime Dynamics Theory Program, RH-192.